

AIRLINE PASSENGER SATISFACTION PREDICTION USING MACHINE LEARNING

A Project Report submitted in partial fulfillment of the requirements for the award of the degree of

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in
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CERTIFICATE



This is to certify that the project titled “**AIRLINE PASSENGER SATISFACTION PREDICTION USING MACHINE LEARNING**” is a bonafide work carried out by **Moulya Lenka (S220165)** , **Niharika Modi (S220168)** in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in the branch of **CSE** during the academic year 2025-2026.

Signature of Guide
Guide Name: **Siva Rama Sastry G**

ABSTRACT

This project focuses on predicting **airline passenger satisfaction** using various **machine learning algorithms**. The dataset includes multiple features such as passenger age, flight distance, in-flight services, and travel class. Data preprocessing techniques such as **data**

cleaning, encoding, and feature scaling were applied to prepare the dataset for model training. Several classification algorithms including **Logistic Regression, Support Vector Machine (SVM), Naive Bayes, Decision Tree, and Random Forest** were implemented and evaluated. The performance of each model was compared using metrics

such as **accuracy, precision, recall, and F1-score**. Among all the models, the **Random Forest Classifier** achieved the highest accuracy, making it the most suitable model for this prediction task. The final model was tested on new input data, and it successfully predicted

whether a customer is **satisfied or dissatisfied**. This project demonstrates how **machine learning** can be effectively used in the airline industry to improve customer experience and decision-making.

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Chapter-1

1 Introduction

This chapter introduces the project background and objectives.

1.1 Background

The rapid growth of the airline industry has led to increased competition among airlines. Customer satisfaction has become a key factor in determining the success of airline services. With the availability of large amounts of passenger data, machine learning techniques can be used to analyze and predict passenger satisfaction effectively.

1.2 Problem Statement

The main problem addressed in this project is to predict whether an airline passenger is satisfied or dissatisfied based on various factors such as flight experience, services, and travel conditions. Accurate prediction helps airlines improve their services and customer experience.

1.3 Objectives

- To analyze airline passenger data.
- To implement different machine learning algorithms.
- To compare model performance using evaluation metrics.
- To identify the best model for prediction.

1.4 Organisation of Report

This report is organized into multiple chapters. Chapter 1 introduces the project. Chapter 2 explains data preprocessing. Chapter 3 covers data visualization. Chapter 4 describes model building and evaluation. Finally, Chapter 5 presents the references used in this project.

Chapter-2

2 Data Preprocessing

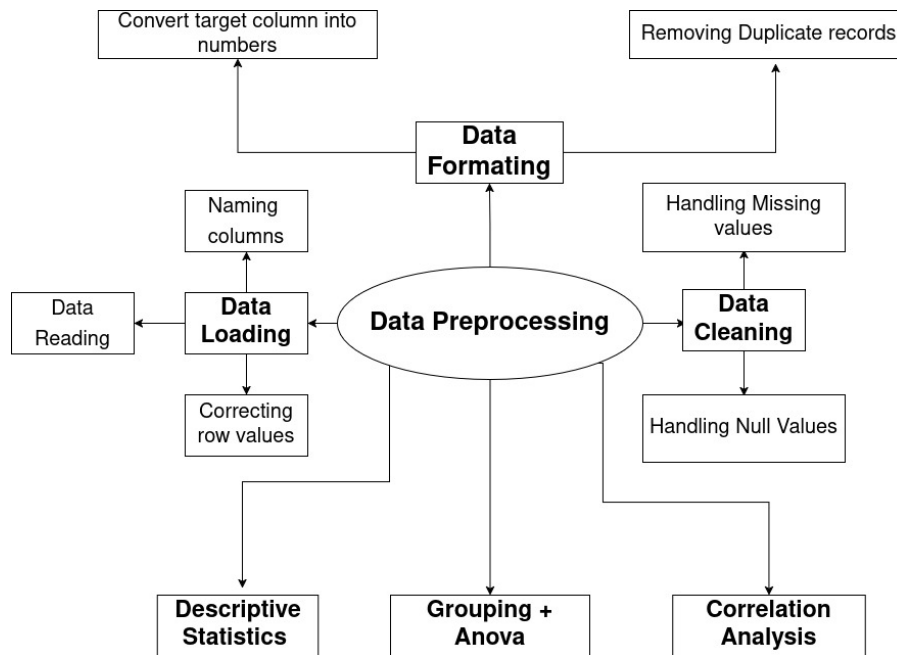


Figure 1: Data Preprocessing Flowchart

Flow Chart Description :-

The flowchart represents the steps involved in the **data preprocessing** phase of the project. It begins with **data loading**, where the dataset is read, unnamed columns are properly labeled, and incorrect row values are corrected.

Next, the process moves to **data cleaning**, which involves handling **missing values** and **null values** to ensure the dataset is complete and consistent. **Duplicate records** are also removed to improve data quality.

After cleaning, the data undergoes **data formatting**, where the **target column is converted into numerical form** for compatibility with machine learning models.

Further, the data is analyzed using techniques such as **descriptive statistics**, **grouping with ANOVA**, and **correlation analysis** to understand relationships between features.

Finally, all these steps collectively prepare the dataset for efficient **model training and prediction**.

Chapter-2

3 Data Visualization

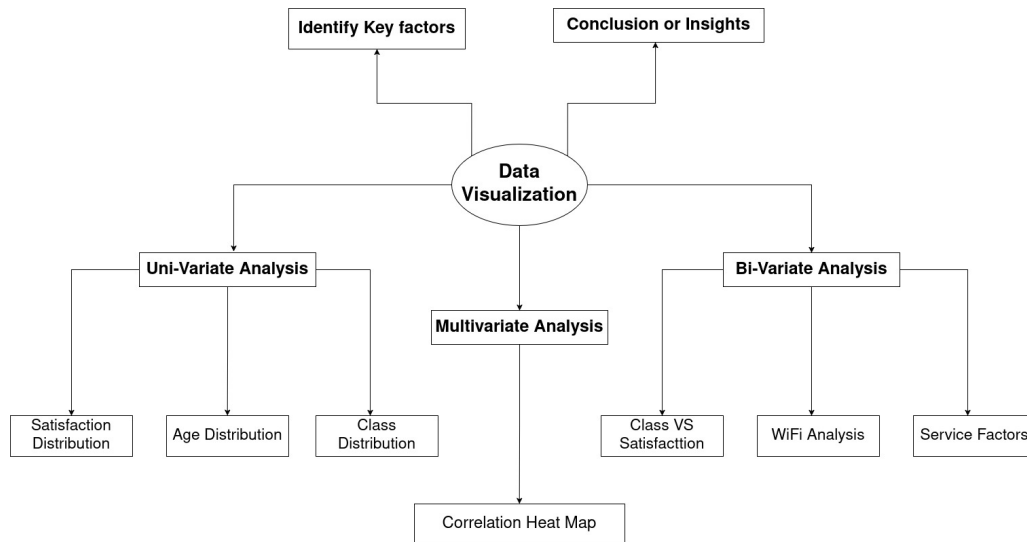


Figure 2: Data Visualization Flowchart

Flow Chart Description :-

The **data visualization** phase plays a crucial role in understanding the dataset by representing it in a graphical format. It helps in identifying patterns, trends, and relationships between different features.

Initially, **univariate analysis** is performed to study individual features. This includes visualization of **satisfaction distribution**, **age groups**, **flight distance**, and **travel class**. These plots help in understanding the overall distribution of customers and their characteristics.

Next, **bivariate analysis** is carried out to examine the relationship between two variables. Important comparisons such as **class vs satisfaction** and service-related factors like **inflight Wi-Fi**, **seat comfort**, and **entertainment** are analyzed. These visualizations highlight how different services impact customer satisfaction.

Further, **multivariate analysis** is performed using techniques like **correlation heatmap** to understand the combined effect of multiple features. This helps in identifying the most influential factors affecting customer satisfaction.

3.1 Visualization Outputs

Univariate Analysis

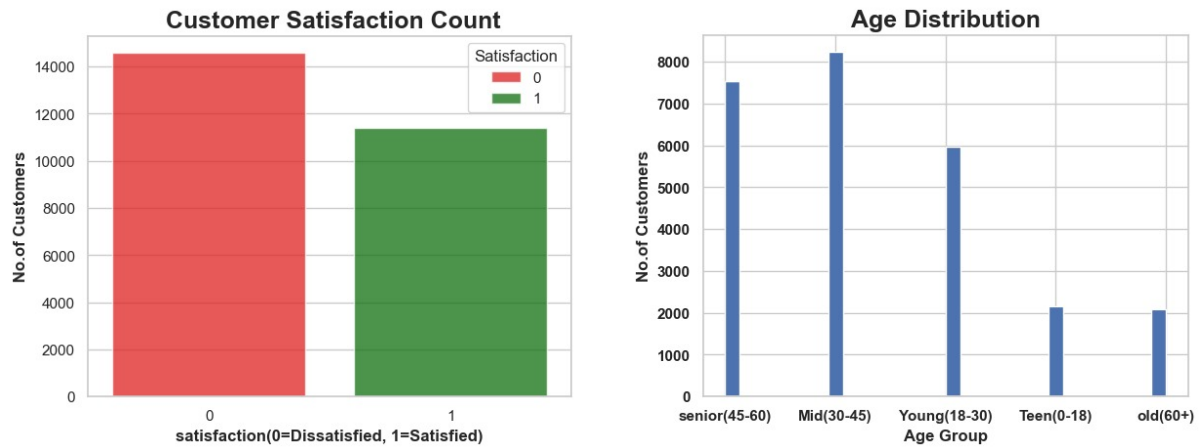


Figure 3: Univariate Analysis Outputs

Bivariate Analysis

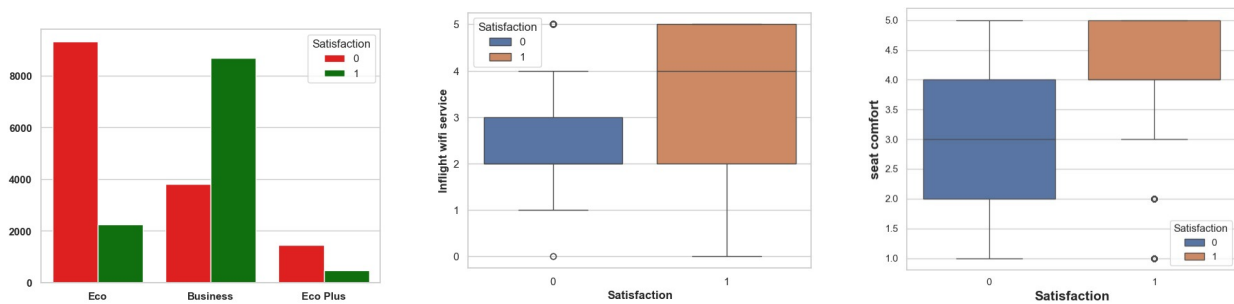


Figure 4: Bivariate Analysis Outputs

Multivariate Analysis

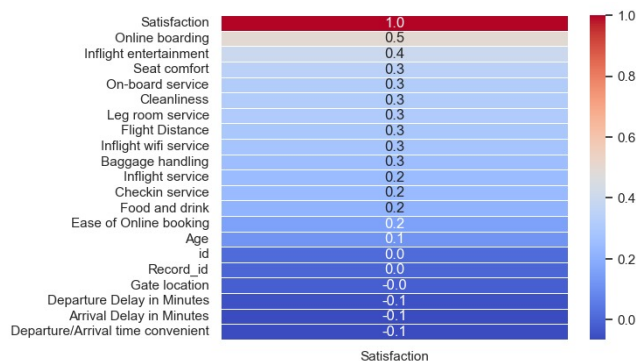


Figure 5: Multivariate Analysis Outputs

Chapter-3

4 Data Modeling

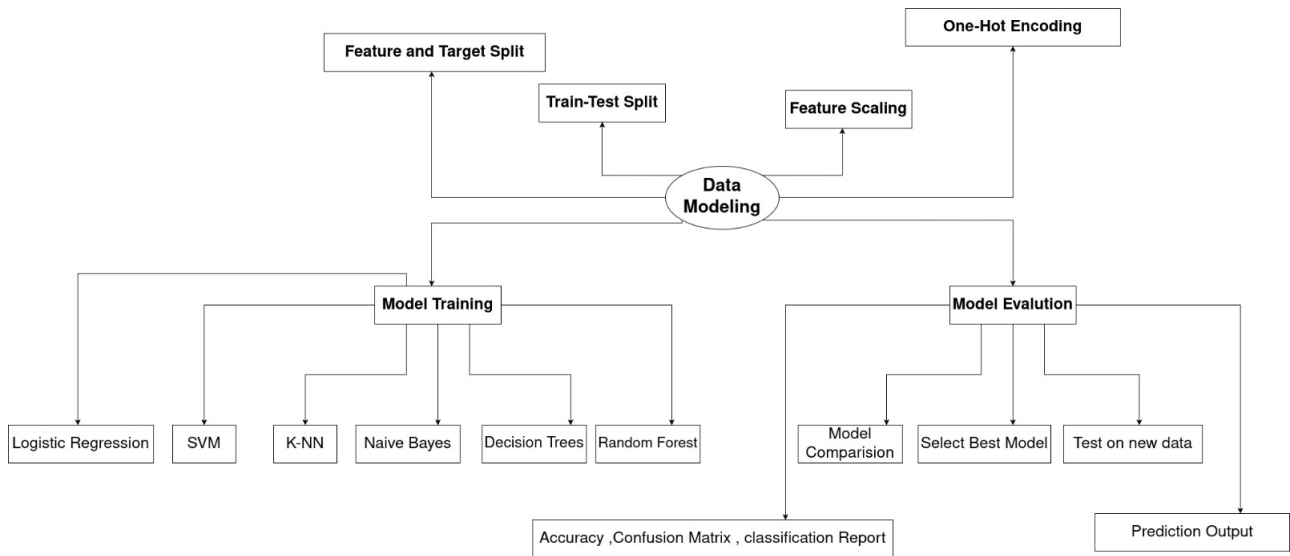


Figure 6: Data Modeling Flowchart

Flow Chart Description :-

The **data modelling phase** focuses on building and evaluating multiple machine learning models to predict customer satisfaction. This phase transforms the processed data into meaningful predictions.

Initially, the dataset is divided into **input features (X)** and the **target variable (y)**, where the target represents customer satisfaction.

Next, **categorical variables** are converted into numerical format using **one-hot encoding**, ensuring compatibility with machine learning algorithms.

The dataset is then split into **training and testing sets** using an 80-20 ratio. The training set is used to build the models, while the testing set is used to evaluate their performance on unseen data.

To improve model performance, **feature scaling** is applied using **StandardScaler**, which standardizes the range of input features.

Several machine learning models are then implemented, including **Logistic Regression**, **Support Vector Machine (SVM)**, **K-Nearest Neighbours (KNN)**, **Naive Bayes**,

Decision Tree, and **Random Forest**.

Each model is evaluated using metrics such as **accuracy**, **confusion matrix**, and **classification report** to compare their performance.

Among all the models, the **Random Forest Classifier** achieved the highest accuracy and demonstrated balanced performance across both classes. Hence, it is selected as the **best-performing model**.

Finally, the selected model is tested on **new customer data**, where it successfully predicts whether a customer is satisfied or dissatisfied along with the prediction probability.

Overall, this phase demonstrates how machine learning models can be effectively used to make accurate predictions and support data-driven decision-making.

4.1 Model Outputs

After training multiple machine learning models, their performance was evaluated using various metrics such as **accuracy**, **confusion matrix**, and **classification report**.

1. Logistic Regression: The model achieved good accuracy, but it was limited in capturing complex relationships in the dataset.

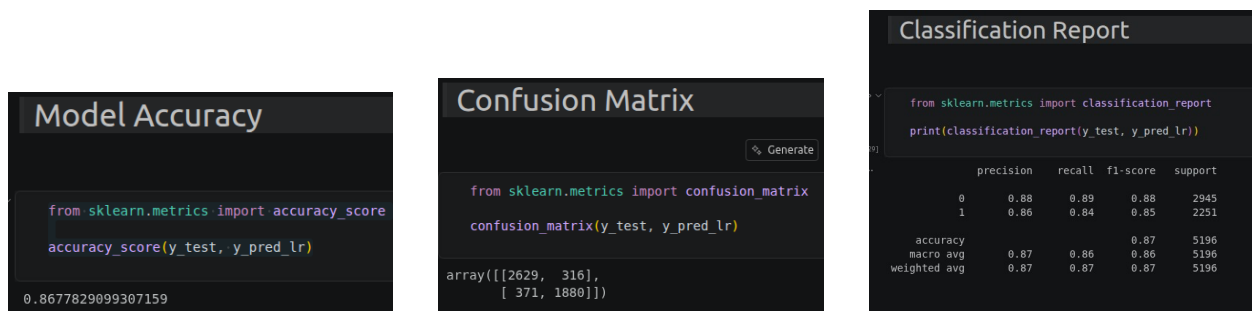


Figure 7: Linear Regression Outputs

2. Support Vector Machine (SVM): The SVM model performed better than Logistic Regression, indicating the presence of non-linear patterns in the data.



Figure 8: SVM Outputs

3.K-Nearest Neighbours (KNN): KNN provided moderate performance but was slightly less effective compared to SVM.

```

Model Accuracy

from sklearn.metrics import accuracy_score

accuracy_score(y_test, y_pred_knn)

0.905504234026174

```

Figure 9: KNN Outputs

4. Naive Bayes: This model showed comparatively lower accuracy, suggesting that it is not suitable for this dataset.

```

Model Accuracy

from sklearn.metrics import accuracy_score

accuracy_score(y_test, y_pred_nb)

0.8314087759815243

```

Figure 10: Naive Bayes Outputs

5. Decision Tree: The Decision Tree model performed well, but it showed slight overfitting and lower generalization compared to ensemble methods.

```

Model Accuracy

from sklearn.metrics import accuracy_score

accuracy_score(y_test, y_pred_dt)

0.9276366435719784

```

```

Confusion Matrix

from sklearn.metrics import confusion_matrix

confusion_matrix(y_test, y_pred_dt)

array([[2750, 195],
       [181, 2070]])

```

```

Classification Report

from sklearn.metrics import classification_report

print(classification_report(y_test, y_pred_dt))

              precision    recall  f1-score   support

     0       0.94      0.93      0.94      2945
     1       0.91      0.92      0.92      2251

 accuracy: 0.93
 macro avg: 0.93      0.93      0.93      5196
 weighted avg: 0.93      0.93      0.93      5196

```

Figure 11: Decision T Outputs

6. Random Forest: Among all models, the **Random Forest Classifier** achieved the highest accuracy of approximately 95%. It also provided balanced performance across both satisfied and dissatisfied classes.

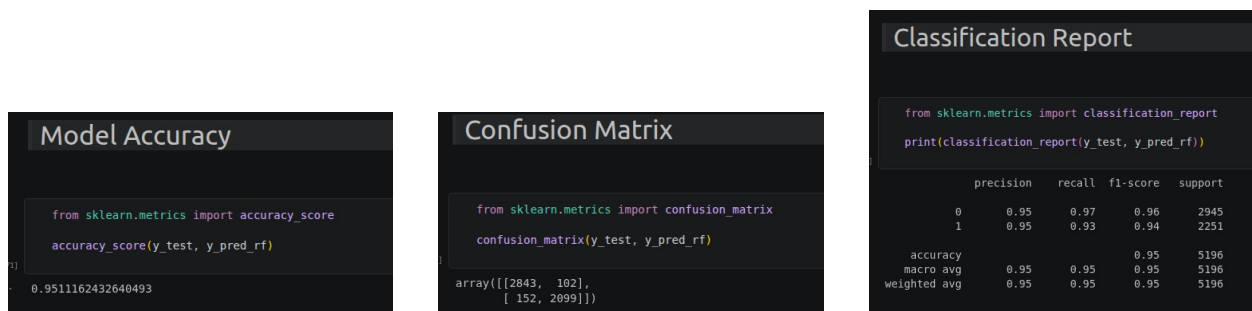


Figure 12: Random Forest Outputs

Best Model Selection: Based on the comparison, **Random Forest** is selected as the best-performing model due to its high accuracy and ability to handle complex feature interactions.

Prediction on New Data: The final model was tested on new customer input data. The model predicted the output as:

- Prediction: **1 (Satisfied Customer)**
- Probability: **77%**

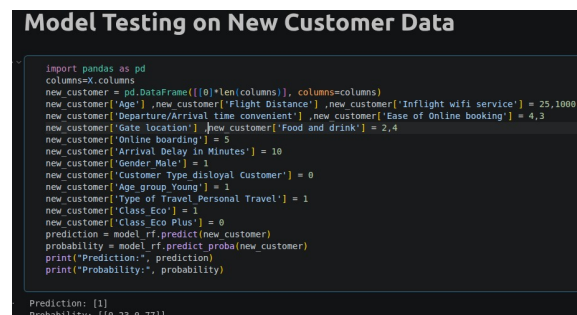


Figure 13: New Data Prediction Outputs

This demonstrates that the trained model can effectively predict customer satisfaction in real-world scenarios.

5 References

- Kaggle Dataset:- https://drive.google.com/file/d/1mUroTEfQ7zKQ6IoR8ox0XY6AtmTtd0zs/view?usp=drive_link/view
- Scikit-learn:- <https://scikit-learn.org/>
- Pandas:- <https://pandas.pydata.org/>
- Seaborn:- <https://seaborn.pydata.org/>
- Seaborn:- <https://matplotlib.org/>